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5. Exercise - 1 - Convolution and Pooling Operations

dimension of output of convolutional layer:

$$W_{out} = \frac{W_{in} - F + 2P}{S} + 1$$

$$H_{out} = \frac{H_{in} - F + 2P}{S} + 1$$

W_{in}, H_{in} = input width & height

F = filter size

S = stride

P = padding

1. Compute the following images output shape

1. Image : 5×5 , Filter : 3×3 , stride : 1, padding = 0 (valid) ↓

→ Since image is a square, width calculating dimensions together.

$$D_{out} = \frac{D_{in} - F + 2P}{S} + 1$$

$$= \frac{5 - 3 + 2 \cdot 0}{1} + 1$$

$$= 3$$

Output shape : $[3, 3]$

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2. Image : 6×6 , Filter : 3×3 , stride : 1, padding : 1^(same)

→ Since Image is a square, calculating dimensions together :

$$D_{out} = \frac{D_{in} - F + 2P + L}{S}$$

$$= \frac{6 - 3 + 2 \cdot 1 + 1}{1}$$

$$= 3$$

Output shape = $[3, 3]$

3. Image : 7×7 , Filter 5×5 , stride : 2, padding : 1

→ since Image is a square, calculating dimensions together :

$$D_{out} = \frac{D_{in} - F + 2P + L}{S}$$

$$= \frac{7 - 5 + 2 \cdot 1 + 1}{2}$$

$$= 3$$

Output shape = $[3, 3]$

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4. Image: 8×8 , filter: 5×5 , stride: 2; padding: 0

→ Since Image is a square, calculating dimensions together.

$$D_{out} = \frac{D_{in} - f + 2P + 1}{s}$$

$$= \frac{8 - 5 + 2 \cdot 0 + 1}{2}$$

$$= 1.5 + 1$$

$$= 2.5$$

Since, Image cannot have a width & height of decimal point, rounding off to lower bound.

Output shape = $[2, 2]$

(same)

5. Image: 10×5 ; Filter: 3×3 ; stride: 1, padding: 1

$$\rightarrow W_{out} = \frac{W_{in} - f + 2P + 1}{s}; H_{out} = \frac{H_{in} - f + 2P + 1}{s}$$

$$= \frac{10 - 3 + 2 \cdot 1 + 1}{1}; \quad = \frac{5 - 3 + 2 \cdot 1 + 1}{1}$$

$$= 9 + 1; \quad = 4 + 1$$

$$= 10$$

Output shape = $[10, 5]$

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2. Compute ^{the} feature map of following:

1. Compute convolution output of following image:

$$I = \begin{bmatrix} 20 & 35 & 35 & 35 & 35 & 20 \\ 29 & 46 & 44 & 42 & 42 & 27 \\ 16 & 25 & 21 & 19 & 19 & 12 \\ 66 & 120 & 116 & 154 & 114 & 62 \\ 74 & 216 & 174 & 252 & 172 & 112 \\ 70 & 210 & 170 & 250 & 170 & 110 \end{bmatrix}_{6 \times 6}$$

$$F = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}_{3 \times 3}, \text{Stride} = 1$$

→ Dimension of output (D_{out}):

$$D_{out} = \frac{D_{in} - F + 2P + 1}{S}$$

$$= \frac{6 - 3 + 2 \cdot 0 + 1}{1}$$

$$= 4$$

Shape of output of convolution = 4

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$$C_{11} = 20 + 35 + 35 + 29 + 44 + 16 + 25 + 21 = 225$$

$$C_{12} = 35 + 35 + 25 + 46 + 42 + 25 + 21 + 19 = 258$$

$$C_{13} = 35 + 35 + 35 + 44 + 42 + 21 + 19 + 19 = 250$$

$$C_{14} = 35 + 35 + 20 + 42 + 27 + 19 + 19 + 12 = 209$$

$$C_{21} = 29 + 46 + 44 + 16 + 21 + 66 + 120 + 116 = 458$$

$$C_{22} = 46 + 44 + 42 + 25 + 19 + 120 + 116 + 154 = 556$$

$$C_{23} = 44 + 42 + 42 + 21 + 19 + 116 + 154 + 114 = 552$$

$$C_{24} = 42 + 42 + 27 + 19 + 19 + 12 + 154 + 14 + 62 = 472$$

$$C_{31} = 16 + 25 + 21 + 66 + 116 + 74 + 216 + 174 = 708$$

$$C_{32} = 25 + 21 + 19 + 120 + 154 + 216 + 174 + 252 = 981$$

$$C_{33} = 21 + 19 + 19 + 116 + 114 + 174 + 252 + 172 = 887$$

$$C_{34} = 19 + 19 + 12 + 154 + 62 + 252 + 172 + 112 = 802$$

$$C_{41} = 66 + 120 + 116 + 74 + 174 + 70 + 210 + 70 = 1000$$

$$C_{42} = 120 + 116 + 154 + 216 + 252 + 210 + 170 + 250 = 1488$$

$$C_{43} = 116 + 154 + 114 + 174 + 172 + 170 + 250 + 170 = 1320$$

$$C_{44} = 154 + 114 + 62 + 252 + 112 + 250 + 170 + 10 = 1224$$

Output of convolution:

225	258	250	209
458	556	552	472
708	981	887	802
1000	1488	1320	1224

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2. Python implementation

3. Apply max pooling to output from 1:

Stride $[2, 2]$

Window Shape $[2, 2]$

→ Dimension of output of max pooling:

$$D_{out} = \frac{D_{conv} - F + 1}{S}$$

$$= \frac{4 - 2 + 1}{2}$$

$$= 1 + 1$$

$$= 2$$

Shape of output of max pooling = $[2, 2]$.

$$\text{Max bet}^n: C_{11}, C_{12}, C_{21}, C_{22} = 566$$

$$\text{Max bet}^n: C_{23}, C_{14}, C_{23}, C_{24} = 552$$

$$\text{Max bet}^n: C_{31}, C_{32}, C_{41}, C_{42} = 1488$$

$$\text{Max bet}^n: C_{33}, C_{34}, C_{43}, C_{44} = 1320$$

$$\text{Max pooling output} = \begin{bmatrix} 566 & 552 \\ 1488 & 1320 \end{bmatrix}$$

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4. Apply a avg. pooling with a 2×2 ^{filter} matrix and stride 2 to following matrix.

$$X = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 13 & 14 & 15 & 16 \end{bmatrix}$$

→ Dimension of output after pooling:

$$D_{out} = \frac{D_{in} - F + 1}{S}$$

$$= \frac{4 - 2}{2} + 1$$

$$= 1 + 1$$

$$= 2$$

Shape of output = $[2, 2]$

$$C_{11} = (1 + 2 + 5 + 6) / 4 = 14 / 4 = 3.5$$

$$C_{12} = (3 + 4 + 7 + 8) / 4 = 22 / 4 = 5.5$$

$$C_{21} = (9 + 10 + 13 + 14) / 4 = 46 / 4 = 11.5$$

$$C_{22} = (11 + 12 + 15 + 16) / 4 = 54 / 4 = 13.5$$

$$\text{Output after avg. pooling} = \begin{bmatrix} 3.5 & 5.5 \\ 11.5 & 13.5 \end{bmatrix}$$

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6. Training a Convolutional Neural Network

6.1. Forward Propagation

TO-DO: Complete the following Convolved Feature Map using stride of 1 and no padding.

			3	3	2	1	0		
0	1	2	0	0	1	3	1	12	
2	2	0	3	1	2	2	3	24	? ?
0	1	2	2	0	0	2	2	?	? ?
			2	0	0	0	1	?	? ?

$$O_{11} = 3 + 6 + 1 + 2 = 12$$

$$\text{dimension of feature map} = \frac{\text{Dim}_{in} - F + 2 \cdot P}{S} + 1$$

$$= \frac{5 - 3 + 2 \cdot 0}{1} + 1$$

$$= 3$$

$$\text{shape of output} = (3, 3)$$

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$$O_{12} = 2 + 2 + 2 + 2 + 4 = 12$$

$$O_{13} = 1 + 2 + 6 + 2 + 6 = 17$$

$$O_{21} = 2 + 6 + 2 = 10$$

$$O_{22} = 1 + 6 + 2 + 4 + 4 = 17$$

$$O_{23} = 3 + 2 + 4 + 4 + 2 + 4 = 19$$

$$O_{31} = 1 + 2 + 4 = 7$$

$$O_{32} = 2 + 4 = 6$$

$$O_{33} = 2 + 6 + 4 + 2 = 14$$

Convolved Feature Map:

12	12	17
10	17	19
7	6	14